

Solution Requirements

Date	4 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users must securely log in or verify their identity to access the HDI prediction dashboard.</i>	<i>High</i>
2	Authorization levels	<i>Define roles (e.g., Admin, Researcher) to control access to specific data and system features.</i>	<i>High</i>
3	External interfaces	<i>Integration with external APIs or datasets to fetch real-time socio-economic metrics.</i>	<i>High</i>

4	Transactions processing	<i>Efficient handling of data inputs and requests between the user interface and the ML engine.</i>	<i>Medium</i>
5	Reporting	<i>Generate and display the predicted HDI score with analytical summaries /insights.</i>	<i>High</i>
6	Business rules, etc.	<i>Apply logic (normalization, data cleaning) to ensure prediction consistency and accuracy.</i>	<i>Medium</i>
7	Compliance to laws or regulations	<i>Ensure data handling adheres to privacy laws and standard research ethical regulations</i>	<i>High</i>
8	<i>Other</i>	<i>Provide a reset/refresh option for users to perform new predictions easily.</i>	<i>Low</i>

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	<i>The system must generate HDI predictions quickly after input.</i>	<i>Response time < 2 seconds for 95% of requests.</i>
2	Scalability	<i>The system should handle increasing numbers of concurrent users.</i>	<i>Support up to 50 concurrent users without performance degradation.</i>
3	Security & Data Privacy	<i>Privacy All user data inputs must be protected against unauthorized access.</i>	<i>100% data encryption (HTTPS/TLS) and secure storage.</i>
4	Reliability & Availability	<i>The application must be accessible and functional at all times</i>	<i>99.9% system uptime availability.</i>
5	Usability & Accessibility	<i>The interface should be easy to use across various devices.</i>	<i>100% responsive design (Mobile, Tablet, Desktop).</i>
6	Other	<i>Compatibility with modern web standards.</i>	<i>Full compatibility with Chrome, Firefox, and Edge browsers.</i>